

This document is an outline which can be used to assist in setting up the ASL3 Allstar image on SHARI Allstar devices equipped with the K4H818 radio module. Kits 4 Hams models using the new K4H818 radio module are known as SHARI RevK models.

When performing an ASL3 setup, some of the advanced setup steps like obtaining the K4H818 utilities, installing Supermon, etc are accomplished using commands typed into the Terminal. The required entries to the terminal line are contained in this document as bold text to assist with this by avoiding typing in long complicated commands, This bold text can be copied (highlight then right mouse click) and then pasted to the terminal line (left click to change the focus and then a right click to paste).

This outline is for latest ASL3 version using the Trixie operating system

To Begin

- Get detailed help from Allstarlink as required <https://allstarlink.github.io/user-guide/pi-detailed/>
- Install the latest Raspberry Pi Imager on your computer [Raspberry Pi software – Raspberry Pi](#)
- (Option) Download the K4H818_update info PDF [K4H818-update app information](#)
- (Option) Download the K4H818_prog info PDF [K4H818-prog app information](#)

Write the ASL3 image to microSD card using Raspberry Pi Imager:

- Plug your microSD card into your computer using a USB adapter or a uSD to SD adapter
- Open Raspberry Pi Imager on your computer
- Select **APP OPTIONS** in the box in lower left corner
- Click on **EDIT** of the **Content Registry**
- Select Use custom URL
- Enter **<https://rpi.allstarlink.org/latest.json>** into the “example” box
- Click on Apply & Restart
- Select your device (Pi5, Pi4, Pi3, Pizero 2W) and click on **NEXT**
- Choose Operating system (AllstarLink) and click on **NEXT**

- Select your storage device and click on NEXT
Hint: If your microSD card does not appear, use <ctrl>+X , then Disk Management. Find your uSD card and right click. Choose Change Drive Letter and Paths then Assign a Drive Letter
- Continue by answering the Customization questions
- Choose Hostname: Recommend <your call> - <your node number> but it is up to you. Just no spaces, use hyphen or underscore
- Localization: Capital City means for the country. For USA it is Washington, D.C. (United States)
- Choose username; **Record and save your Hostname, username and password**, lest you forget them
- Choose WiFi; Configure wireless LAN if desired then click on NEXT
- SSH authentication: Enable it
- Click on **WRITE**
- Click on **I UNDERSTAND, ERASE AND WRITE**
- Click on **FINISH**

Configure Node:

- Remove microSD card and insert in Pi
- Plug in ethernet cable if not using WiFi
- Apply power to the Pi
- Wait for the green HB LED on the SHARI board to begin blinking (not the green LED on the Pi)

Log in:

(You use your browser (Edge, Chrome, etc) to connect to your node)

- Open a new tab in browser
- Enter your Hostname followed by dot local (i.e. <hostname>.local) into address bar
- Click on Advanced and the Continue to.....(unsafe)
- The cockpit opens – click on the Web Admin Portal picture
- Reports as “Your connection isn’t private”– proceed anyway by clicking on Advanced
- Click on Continue to(unsafe)
- Enter User Name and Password The one’s you entered into Raspberry Pi Imager (above in yellow)

- Click on **Turn on administrative access** when the dashboard opens. Enter your password again as requested.
- Click on Overview on the left side. **If Useage for the CPU is showing more that 2% activity, sit back and wait until it is less than 2%. It is still updating. If updates are available, the node may automatically "reboot" after the initial boot, initial package updates, and initial configuration updates so you will see a "Reconnect" message when it is done with this initialization**
- Click on Reconnect and Log in again if necessary

Updates:

If this is a new install there should be no updates. To check, click on Software updates to see and install if necessary

Note: Click on Terminal and enter **sudo reboot** after any updates (unless you switched Reboot to on)

Set up your node:

- Click on Terminal (menu column on the left)
- Enter **sudo asl-menu** (Note: First time using sudo requires you to enter your log-in password)
Use the keyboard up/down arrow keys to move between menu choices
Use the TAB key to move between the menu and choices at the bottom
- Highlight 1 Node Settings <enter> → 1 then Allstar Node Setup Menu <enter>
- Enter your node number from Allstarlink.org
- Enter node password (**Note: This is from Allstarlink.org Portal>Node Settings, not the password you use for ASL3**)
- Highlight Default/manual configuration (show all settings) and press <enter>

Note: You are given a choice of Default/manual configuration, Common Configurations and Device Configurations. This outline is using Default Configurations so that you become more aware of the settings involved in the setup. Feel free to choose the other two selections at any time.

- 3 Node callsign <enter> → Enter your callsign
- 4 Radio interface → SimpleUSB
- 5 Duplex type <enter> → Choose: 1 Half-duplex node (Hotspot) with telemetry tones
Note: Up/Down arrow keys move red box. Space bar confirms choice (moves the asterisk)
- 6 Post node status set to Yes

- 8 Interface Tune CLI → Change/set the following selections after restarting Asterisk

Note: For the K4H818 we do not change selection J) as we do for the SA818

- Restart Asterisk now
 - 2) RX Voice Level = 650 Note: This requires several <enters> to accomplish)
 - 3) TX Channel A=600
 - G) Toggle PL Filter=disable
 - W) Write
 - 0) Exit Menu
- Push <tab> twice to get to <back> and press <enter>
 - 4 Update Asterisk IAX port if it is not 4569
 - (If this is your first node use 4569. Go up, i.e. 4570 or down 4568 etc for additional node numbers)
 - 3 Update Asterisk AMI Password (you can change this or use the default)] Note: copy it and save for installing Supermon later in this outline
 - 2 Restart Asterisk if 3 or 4 changed
 - Do a <tab>,< tab to get to <back> and press <enter>
 - Go to Terminal using <tab>,< tab>, <Exit Main Menu>,<Yes>

Setup and Program the K4H818:

The K4H818 does not use the ASL3 SA818 setup menu (sudo sa818-menu) to set up the radio frequency PL, squelch, etc. For the K4H818, we install several programs and use them to setup the radio and flash new firmware. At the Terminal:

- Enter **wget -O - https://raw.githubusercontent.com/Kits4Hams/K4H818_utilities/main/install_asl3.sh | bash**
This installs K4H818 utility programs in the /home/<user name> directory and a program called AVRdude (if not already installed) used to reflash the K4H818. The installation process will end with the words:

Done. Run with:

/home/<your ASL3 user name>/K4H818-prog.py

/home/<your ASL3 user name>/K4H818-update.py

Re-run this same installer any time to update both to the latest version and re-check avrdude.

- Enter **cd /home/<your ASL3 user name>**
- Enter **ls -l** Note: the l is lower case L
- Verify the following files are shown:
K4H818_example.ini
K4H818-prog.py
K4H818-update.py
- Enter **./K4H818-update.py** This flashes the latest firmware to the K4H818
- Enter **./K4H818-prog.py** This allows you to setup the K4H818

Hint Use **K4H818-prog -d** to use pre-selected configurations. Use **<ctrl>+c** to exit K4H818-prog and start over

The K4H818 is rebooted and locked into the setup mode. It will not respond to HT RF inputs while running K4H818-prog. The current configuration as read from the K4H818 is presented. Then an interactive session is started. Each new question presents the current setting which can be selected by pressing <enter>. There are 5 subaudible tone types possible. We recommend using a subaudible tone (choices 1 thru 4). Squelch can be set from 0 (unsquelched) to 8. We recommend an RX volume setting of 30. We recommend all filters be enabled (i.e. y) Here is a sample output.

Current configuration read from the K4H818:

Firmware Version: 1.5.13
Tx Frequency: 435.000 MHz
Rx Frequency: 435.000 MHz
Tx Sub-audible Type: CTCSS
Tx CTCSS Frequency: 100.0 Hz

Rx Sub-audible Type: CTCSS
Rx CTCSS Frequency: 100.0 Hz
Squelch Value: 5
RX Volume: 30
Pre/De-Emphasis Enabled: y
Voice High Pass Enabled: y
Voice Low Pass Enabled: y
CTCSS/CDCSS High Pass Enabled: y
CTCSS/CDCSS Low Pass Enabled: y
RSSI Low Pass Filter Enabled: y

COS LED:

The K4H818 drives the COS (yellow) and CSS (blue) LEDs directly so no setup is required for the SHARI RevK

Note: CSS means CDCSS (i.e.PL) or CDCSS (i.e. digital PL)

Verify/Adjust RX Voice Level:

- Set up a radio to the frequency and subaudible tone you just programmed into the K4H818
- Key up and verify the yellow COS and CSS LED come on.
- Unkey and verify the red PTT LED lights while the node transmits a courtesy tone.
- Key up and connect to the Parrot node (55553) by dialing DTMF sequence *355553 and unkey
 - After connection message key up and ID with your call and a short voice message. Unkey and listen for the comment on level followed by a parrot of your voice message. If the level is not correct, change the distance of your mouth to the microphone on the radio, change the mic gain on your radio or go to 8 Interface Tune CLI (**sudo asl-menu**) and change 2) RX Voice Level = 650 to a different value. You can just keep adjusting the level from 650 until you have it correct. Dial *10 to disconnect from the parrot node

WiFi:

Set up the correct country code for WiFi (**Failure to do this will render WiFi inoperable**)

- Enter **sudo raspi-config**
- Choose *5 Localisation Options*
- Choose *L4 WLAN Country*
- Select the country in which the Pi is to be used (US for USA)
- <OK> then <Finish> to return to Terminal

If you want WiFi and did not set it up in the data entered into Raspberry Pi Imager

- Go to WiFi manager
- Scan for WiFi networks (if necessary)
- Enter SSID and Passphrase of desired WiFi connection(s)
- Return to terminal

Install Supermon:

From the Terminal (Note: A more detailed version is available at <https://tinyurl.com/Supermon-NEW>)

- **cd /usr/local/sbin**
- **sudo wget "http://2577.nodes.allstarlink.org:43856/supermonASL_fresh_install" -O supermonASL_fresh_install**
- **sudo chmod +x supermonASL_fresh_install**
- **hash**
- **sudo supermonASL_fresh_install**

Note: This take a minute or more to execute. **Manually scroll the terminal if auto-scrolling stops**

- **sudo wget "http://2577.nodes.allstarlink.org:43856/supermonASL3_latest_update" -O supermonASL3_latest_update**
- **sudo chmod +x supermonASL3_latest_update**
- **hash**
- **sudo supermonASL3_latest_update**

Edit the allmon.ini file using the Nano editor

- **cd /var/www/html/supermon**

- **sudo nano allmon.ini**
 - Change nodes=12345,12346 to nodes=<your node number>
 - Change [12345] stanza to <your node number>
 - Change passwd=secret to passwd=<the AMI password you were told to copy and save earlier in the outline>
 - Comment out (start each line with a semicolon ';') the [12346] stanza heading and each line in the stanza
Note: This may already be done with the latest Supermon version
 - In the [LsNodes] stanza changenode=12345 tonode=<your node number>
 - Exit and save with Ctrl>+X, then Y (Yes) then <enter>

Setup the User and Password for you to log into Supermon

- **sudo rm .htpasswd**
- **sudo htpasswd -cB .htpasswd admin** (if a different user name than admin is desired change it here)

Hint: You will be asked to enter the password twice, typing of the password does not show on the screen

Install Allmon3:

- **cd /**
- **sudo allmon3-passwd --delete allmon3** (error message is OK)
- **sudo allmon3-passwd admin** (if a different user name than admin is desired change it here)
- **sudo reboot**
- Wait for the blinking green HB LED on SHARI RevK, then Reconnect

Verify Allmon3:

- open new tab in browser
- URL = <host name>.local/allmon3
- verify Allmon3 login works

Finish Supermon Setup:

- open new tab in browser
- <host name>.local/supermon
- click on Nodes, then click on your node number to Activate the node number (**must do this to log in**)
- click on Login then login using Tab key to navigate - not the mouse
- Click on “Configuration Editor” button and select global.inc to edit. You can change anything in quotes.
- Recommend:
 - \$CALL = "Your Callsign";
 - \$NAME = "Your Name";
 - \$LOCATION = "Howell, MI";
 - \$TITLE2 = "12345 Cloud Server"; (we recommend a description of your node. i.e. SHARI PiHatU NeoPi5)
 - \$TITLE3 = "AllStarLink/IRLP/EchoLink/Digital - Bridging Control Center"; (we recommend “Node <your node number>”)
 - \$BACKGROUND = "background.jpg"; (we recommend deleting background.jpg so you have two quotes i.e. “”)
 - \$BACKGROUND_COLOR = "green";
 - \$SMSERVERNAME = "12345 Cloud"; (we recommend Supermon <your node number>)
 - \$LOCALZIP = "48843"; (your zip code to get your local weather)
 - \$DVM_URL_NAME = "Kits 4 Hams";
 - \$DVM_URL = "http://kits4hams.com"; (also delete the “//” at the beginning of the line)
- Click on “Write your Edits” box at bottom to save your edits
- Click on Return to Index
- Click on Close Window
- Refresh your browser window

Backup:

- **sudo asl-menu**
- B Backup and restore Menu
- 1 Create node backup

Shutdown

- Back to Overview
- Shutdown (top right corner) or **sudo shutdown -h now** at the terminal